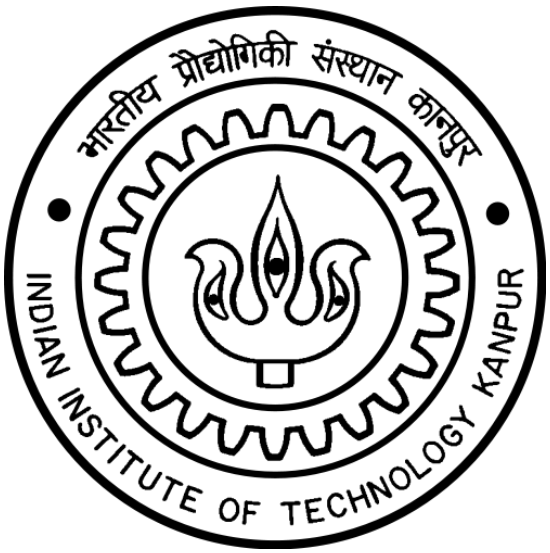
INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Department of Electrical Engineering



Quantum Tunneling in Relativistic Systems

SURGE Project Report

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Preface

Graphene is a mono-layer of graphite, an allotrope of carbon just like diamond. While graphene was theorized in 1947 by Philip R. Wallace, it was isolated over 50 years later in 2004 by Andre Geim and Konstantin Novoselov. They were subsequently awarded the Nobel Prize in 2010 for their groundbreaking work. Graphene has become a central part in the study of condensed matter physics since its discovery because it is a playground for several interesting and counterintuitive phenomena. In this report, we will explore some of these phenomena in depth and talk about their promising future aspects. This will begin with a survey of the basic principles and groundwork we must cover in order to understand these phenomena, which will be covered in Chapter 1. In Chapter 2, we shall talk about a classic phenomenon predicted well before graphene was even theorized, called the Klein Paradox or Klein Tunneling. In Chapter 3, we shall explore another peculiarity that shows up in graphene, called the Veselago lens. In Chapter 4, we shall talk about some of the future directions that this work could take looking forward.

Acknowledgements

I would like to thank my mentor, Prof. Amit Agarwal, for giving me the opportunity to pursue this project and also for the guidance he gave me along the way. He placed great emphasis on original thinking, learning physics by not just reading it from a paper or a book, but understanding its history and how these discoveries were made and also on understanding things mathematically and inferring the physical insights from those results. I would then like to thank my parents for the continued support I received during this period. Summers at IIT Kanpur are not exactly described by the word "pleasant" and I got sick several times and ended up losing a lot of precious time I could have devoted to this project. My parents and friends ended up helping me quite a bit through this challenging period and I couldn't be more grateful.

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Chapter 1

Preliminaries

1.1 Dirac's Equation in 3+1 Dimensions

Schrödinger's equation is not consistent with the special theory of relativity. This is easily deduced owing to the fact that space and time are not treated equally, i.e. there is only a single derivative with respect to time, while we have double derivatives with respect to the spatial co-ordinates. Therefore, a search for an equation for relativistic particles began. The answer was provided by Dirac. First, let's write the energy dispersion for a classical relativistic particle.

$$E^2 = p^2c^2 + m^2c^4 \tag{1.1}$$

Let's take the square root of this equation an assume that it will give us a linear function of the momentum.

$$p^2c^2 + m^2c^4 = (\alpha_0 mc^2 + \alpha_1\hat{p}_1 + \alpha_2\hat{p}_2 + \alpha_3\hat{p}_3)^2 \tag{1.2}$$

$$\implies \alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}. \tag{1.3}$$

$\{\alpha_i\}$ are not numbers - they are matrices. As it turns out, they have to be at least 4×4 matrices (higher dimensions don't give any new information and lower dimensional matrices can't satisfy the above relations. We can explicitly find these 4 matrices.

$$\boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \tag{1.4}$$

where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ and $\beta \equiv \alpha_0$. I is the 2×2 identity matrix and $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ are the Pauli matrices.

Thus, we can take the square root and obtain the Dirac equation by replacing E and p

with their respective quantum mechanical operators.

$$i\hbar\frac{\partial\psi}{\partial t} = (-i\hbar c\boldsymbol{\alpha}\cdot\nabla + \beta mc^2 + V)\psi \tag{1.5}$$

where V is the potential. It's a bit strange that there are 4 components in the wavefunction instead of 2 (which would describe a spin-$\frac{1}{2}$ particle). It turns out that two of the components are related to negative energy solutions. If we solve the problem of a free Dirac particle, we obtain the following eigen-energies

$$E = \pm\sqrt{m^2c^4 + p^2c^2} \tag{1.6}$$

The negative energy solutions are a problem here since electrons should be able to emit photons and reach lower energies, with no bounds. But, we do observe positive energy electrons. We also can't discard these solutions since the description of a general wavefunction needs the eigenvectors associated with these negative energy solutions. Dirac proposed that the negative energy states are already all occupied. What we call the vacuum is actually the occupied and unobservable sea of negative energy electrons. If this were true, then the electrons couldn't end up in these negative energy states owing to Pauli's exclusion principle. This picture also predicts antiparticles. These antiparticles are called positrons in this case and it can be shows that they have the opposite charge of an electron, but the same mass. There are issues with this picture because if all the negative energy states are occupied, then the vacuum has infinite negative energy and also infinite negative charge. This was rectified by Quantum Field Theory later on, but that is outside of the scope of this project, so we shall accept Dirac's proposition.

1.2 Dirac's Equation in 2+1 Dimensions

With only 2 spatial dimensions, we can write

$$E^2 = (p_x^2 + p_y^2)c^2 + m^2c^4 = (\alpha_1 p_x + \alpha_2 p_y + \alpha_0 mc^2)^2 \tag{1.7}$$

which, on comparing coefficients, implies

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij} \quad \text{where } i,j \in \{0,1,2\} \tag{1.8}$$

Again, the coefficients can not be complex numbers. The difference between this and the 3+1 dimensional case is that here, we need only 3 matrices that obey the anti-

commutation relations. The matrices can be chosen to be the Pauli matrices.

$$\alpha_1 = \sigma_x, \quad \alpha_2 = \sigma_y, \quad \alpha_0 = \sigma_z \tag{1.9}$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{1.10}$$

So, the 2+1 dimensional Dirac equation can be written as

$$i\hbar \frac{\partial \psi}{\partial t} = (c\boldsymbol{\sigma} \cdot \hat{\boldsymbol{p}} + \sigma_z mc^2)\psi \tag{1.11}$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ and $\boldsymbol{p} = (p_x, p_y)$.
The primary thing to note here is that the wavefunction only contains two components. This time, the second component is associated with negative energy solutions, as can be seen by following the same steps that we did for the 3+1 dimensional case. Spin is not encoded into this equation, so one has to put it in by hand.

1.3 Band Structure of Graphene

The connection between this and the previous section will become clear at the end of this chapter.
The tight binding model can be thought of as just a discretization of space. If we have regularly arranged lattice points and we wish to find the wavefunction, then we only care about its values at the lattice points since the lattice separation is our new smallest distance. In discrete space, we replace derivatives with differences. We shall take the nearest neighbour approximation, where these derivatives are only replaced by nearest neighbour differences. As an example, if we have a straight chain of lattice points,

$$\psi' \to \frac{\psi_{i+1} - \psi_i}{a} \tag{1.12}$$

where a is the lattice separation or simply the lattice constant and ψ_i is the amplitude of the wavefunction at the ith lattice point.
We need to find the laplacian operator for graphene, which is the sum of double derivatives because it appears in the kinetic energy term of the Schrödinger equation. The electrons are considered free, so the potential is zero. It should be pointed out that graphene is divided into 2 sublattices as shown in Figure 1. We shall refer to red dots as sublattice B and the white dots as sublattice A. To write the derivatives, we will use the concept

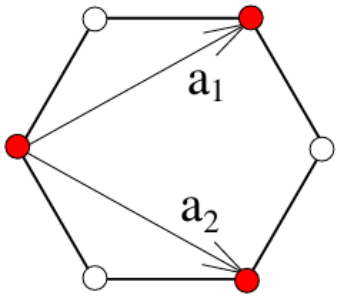


Figure 1.1: Hexagonal Lattice

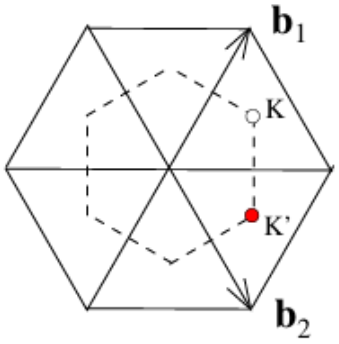


Figure 1.2: Reciprocal Lattice of Graphene

of directional derivatives. In Figure 1.1, take the right direction as the positive x axis and the upwards direction as the positive y axis. Label the wavefunction amplitudes at the white points as $\psi_{A,R}$ and the red point to its immediate right as $\psi_{B,R}$. These pair of amplitudes for all values of the vector R will tell us the probability amplitudes at all points on the lattice (x and y subscripts denote partial derivatives). Note that the lattice constant is taken to be 1 to simplify the equations.

$$\psi_x^+ \equiv \psi_{B,R} - \psi_{A,R} \tag{1.13}$$

$$\frac{1}{2}\psi_x^- - \frac{\sqrt{3}}{2}\psi_y^+ \equiv \psi_{A,R} - \psi_{B,R-a_2} \tag{1.14}$$

$$\frac{1}{2}\psi_x^- + \frac{\sqrt{3}}{2}\psi_y^- \equiv \psi_{A,R} - \psi_{B,R-a_1} \tag{1.15}$$

The $+$ and $-$ superscripts denote the right hand and left hand derivatives. These equations are the result of writing derivatives for the 3 nearest neighbours.

$$\nabla^2\psi \equiv \frac{\psi_x^+ - \psi_x^-}{\frac{3}{2}} + \frac{\psi_y^+ - \psi_y^-}{\sqrt{3}} \tag{1.16}$$

The right hand side expression can be obtained by subtracting equations (1.14) and (1.15) from (1.13).

$$\nabla^2\psi = \frac{2}{3}(\psi_{B,R} + \psi_{B,R-a_1} + \psi_{B,R-a_2} - 3\psi_{A,R}) \tag{1.17}$$

Define $t \equiv \frac{\hbar^2}{3m}$. Then, we can write Schrödinger's equation as

$$-t(\psi_{B,R} + \psi_{B,R-a_1} + \psi_{B,R-a_2} - 3\psi_{A,R}) = E\psi_{A,R} \tag{1.18}$$

Similarly, writing derivatives around a point from sublattice B, we get

$$-t(\psi_{A,R} + \psi_{A,R+a_1} + \psi_{A,R+a_2} - 3\psi_{B,R}) = E\psi_{B,R} \tag{1.19}$$

We might as well get rid of the term with coefficient $3t$ since it just serves to shift the energies up and down. Our ansatz for the above equations is

$$\begin{pmatrix} \psi_{A,R} \\ \psi_{B,R} \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \tag{1.20}$$

$$\implies E \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} 0 & \gamma(k) \\ \gamma^*(k) & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \tag{1.21}$$

where $\gamma(k) = t(1 + e^{i\boldsymbol{k}\cdot\boldsymbol{a_1}} + e^{i\boldsymbol{k}\cdot\boldsymbol{a_2}})$. Finally, we find

$$E(\boldsymbol{k}) = \pm\sqrt{|\gamma(k)|^2} = \pm t\sqrt{1 + 4\cos(\frac{3k_x a}{2})\cos(\frac{\sqrt{3}k_y a}{2}) + 4\cos^2\left(\frac{\sqrt{3}k_y a}{2}\right)} \tag{1.22}$$

where we've reinserted a using dimensional analysis. In Figure 1.2, the dotted hexagon inside is the Wigner-Seitz cell of the reciprocal lattice or the first Brillouin zone. It is labelled by two point K and K' which are the two Dirac points. The other 4 corners of the first Brillouin zone are equivalent to either K or K' because they are connected by reciprocal lattice vectors to one of the Dirac points. If we expand the energy dispersion obtained about K, we get something surprising.

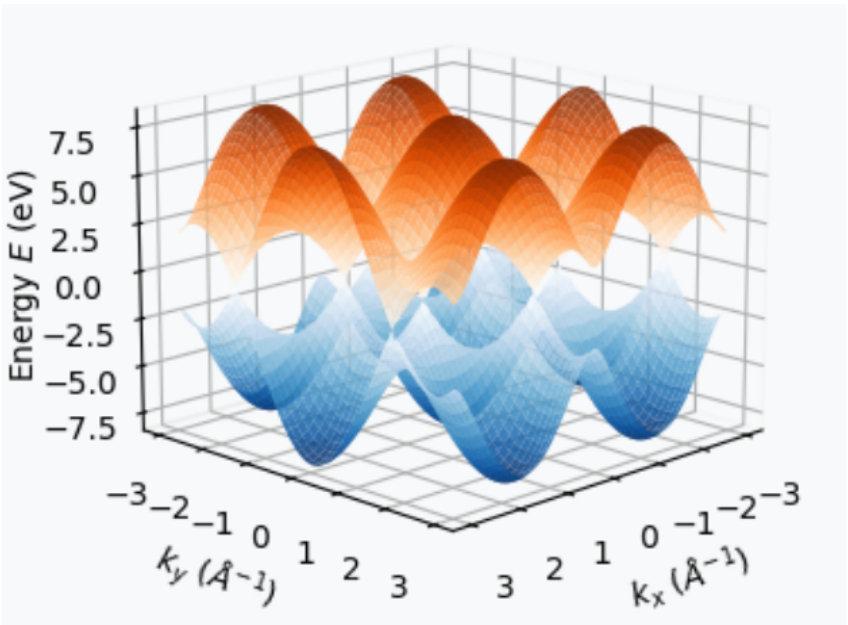


Figure 1.3: Band Structure of Graphene

$$E(\boldsymbol{q}) = \pm\hbar v_F |\boldsymbol{q}| \tag{1.23}$$

where $\boldsymbol{q}$ is a small deviation from the wavevector $\boldsymbol{K}$ and $v_F = \frac{3a|t|}{2\hbar}$ is the fermi velocity. This linear dispersion can be mapped to the Dirac equation when the mass of the electrons is 0 and the speed of light is replaced by the fermi velocity.
The Hamiltonian can be expressed as

$$\mathcal{H} = \hbar v_F \boldsymbol{\sigma} \cdot \boldsymbol{q} \tag{1.24}$$

$$\implies \mathcal{H} = -i\hbar v_F \boldsymbol{\sigma} \cdot \nabla \tag{1.25}$$

Expanding about K', we obtain

$$\mathcal{H}' = \mathcal{H}^T \tag{1.26}$$

The wavefunction will actually contain four components - 2 corresponding to the sublattice degree of freedom and two corresponding to the two Dirac points. So, an electron with energy E can live in 4 linearly independent states.

Chapter 2

Klein Tunneling

2.1 Creation of Potential Barriers

We now look at the problem of tunneling in relativistic quantum mechanics in the context of graphene. The first question to ask is how a potential barrier is created in graphene. The answer is provided by an N-P-N junction. In the N sides, let the energy of the electron be E.

$$E = \hbar v_F k_F \tag{2.1}$$

where $\boldsymbol{k_F}$ is the fermi wavevector. We know from basic semiconductor physics that in a 2-D solid

$$k_F = \sqrt{2\pi n} \tag{2.2}$$

where n is the electron concentration in the N region. However, we know from the previous chapter that a wavefunction has 4 components, which means a particle with energy E can occupy 4 different states. The actual expression then becomes

$$k_F = \sqrt{\pi n} \tag{2.3}$$

In the P region, the fermi wavevector will be $q_F = \sqrt{\pi p}$. For $p > n$, $q_F > k_F$, which means that the energy is higher in the P region. It stands to reason that this extra energy must be coming from a potential of some kind given that the energy of the electron remains the constant throughout. So, the way to think about potential barriers is N-P-N junctions.

2.2 Klein Paradox

Consider a potential of the kind

$$V(x) = \begin{cases} V_0, & 0 < x < D, \\ 0, & \text{else.} \end{cases} \tag{2.4}$$

where D is the width of barrier. Solving this problem is straightforward. The Hamiltonian is given by

$$\mathcal{H} = i\hbar v_F \boldsymbol{\sigma} \cdot \nabla + V(x) \tag{2.5}$$

Our ansatz for an electron incident at angle ϕ with respect to the x axis is

$$\psi_1(x,y) = \begin{cases} \left(e^{ik_x x} + re^{-ik_x x}\right) e^{ik_y y}, & x < 0, \\ \left(ae^{iq_x x} + be^{-iq_x x}\right) e^{ik_y y}, & 0 < x < D, \\ te^{ik_x x + ik_y y}, & x > D, \end{cases} \tag{2.6}$$

$$\psi_2(x,y) = \begin{cases} s\left(e^{ik_x x + i\phi} - re^{-ik_x x - i\phi}\right) e^{ik_y y}, & x < 0, \\ s'\left(ae^{iq_x x + i\theta} - be^{-iq_x x - i\theta}\right) e^{ik_y y}, & 0 < x < D, \\ ste^{ik_x x + ik_y y + i\phi}, & x > D. \end{cases} \tag{2.7}$$

where ψ_1 and ψ_2 are the components of the same wave function, $k_x = k_F \cos\phi$, $k_y = k_F \sin\phi$, $q_x = \sqrt{\frac{(E-V_0)^2}{\hbar^2 v_F^2} - k_y^2}$. $\theta = \tan^{-1} k_y/q_x$, $s = \operatorname{sgn}(E)$ and $s' = \operatorname{sgn}(E - V_0)$. Just like non-relativistic quantum mechanics, to find r and t, we impose the constraint of continuity on the wavefunction since the discontinuity in the potential is finite. Note that the derivative may not be continuous, as that condition was a result of integrating Schrödinger's equation, which was second order in space unlike Dirac's equation.

$$r = \frac{2e^{i\phi}\sin(q_x D)\left(\sin\phi - ss'\sin\theta\right)}{ss'\left[e^{-iq_x D}\cos(\phi+\theta) + e^{iq_x D}\cos(\phi-\theta)\right] - 2i\sin(q_x D)} \tag{2.8}$$

The transmission probability is then given by $T = 1 - |r|^2$ for probability to be conserved. Immediately obvious is the fact that if $q_x D = n\pi$, where n is an integer, then $r = 0$ and therefore $T = 1$. This resonance condition is related to the length and wavevector perpendicular to the barrier, not the actual length travelled or the fermi wavevector inside the barrier. This is analogous to the resonance condition one gets if we consider

a non relativistic 2D tunneling problem with the same potential with $E > V_0$, since the solution in the barrier are also plane waves in that case. However, the biggest consequence of equation (2.8) is that if $\phi = 0$, then $r = 0$ and $T = 1$ and this holds regardless of the height and width of the barrier, which is very counterintuitive even compared to non relativistic quantum mechanics because at least there, the probability reduced as the width and height of the barrier grew. This is related to the fact that the incident spinor and reflected spinor in the case of normal incidence are orthogonal to each other, i.e. the pseudospins of the incident and reflected waves are orthogonal. The potential cannot affect the pseudospin orientation since it is a multiple of the identity operator. So, the incoming wave must tunnel through with probability 1.

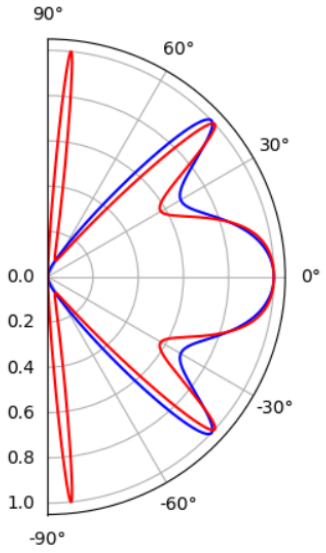


Figure 2.1: Polar plot of transmission probability as a function of ϕ with $D = 100\ nm$, $n = 10^{16}\ m^{-2}$, $p = 3 \times 10^{16}\ m^{-2}$ (red) and $p = 5 \times 10^{16}\ m^{-2}$ (blue)

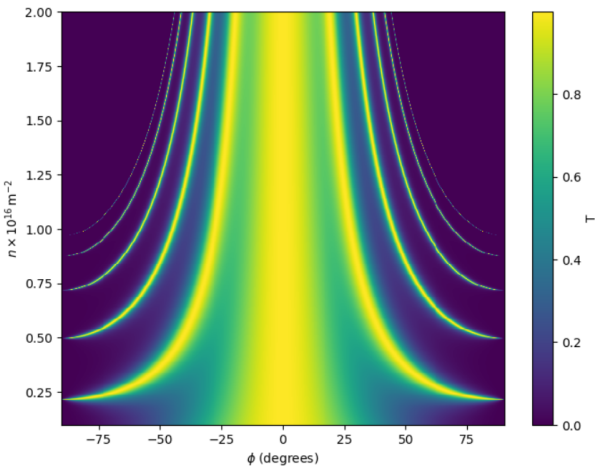


Figure 2.2: Color plot of transmission probability as a function of incidence angle and electron concentration in the n-side with $D = 100\ nm$ and $p = 10^{16}\ m^{-2}$

Chapter 3

Veselago Lens

3.1 PN Junction in Graphene

We have discussed everything in terms of wavevectors thus far, but the actual direction of motion is determined the group velocity. In the N side, the group velocity is actually parallel to the wavevector.

$$v_g = \frac{1}{\hbar}\frac{\partial E}{\partial \boldsymbol{k}} = v_F \frac{\boldsymbol{k}}{|\boldsymbol{k}|} \tag{3.1}$$

So, v_g is parallel to $\boldsymbol{k}$ in the N region. In the P region,

$$v_g = \frac{1}{\hbar}\frac{\partial E}{\partial \boldsymbol{q}} = -v_F \frac{\boldsymbol{q}}{|\boldsymbol{q}|} \tag{3.2}$$

v_g is not parallel to $\boldsymbol{k}$. Now, if our wave has to travel to the right, the group velocity must always point towards the positive x axis, i.e. the x component of the group velocity must be positive. So, in the P region, the wavevector's x component must be negative, i.e. $q_x < 0$. The y component of the wavevector will remain conserved because the system is symmetric under translations along the y direction. The trajectories of electrons emanating from a source are displayed in Figure (3.1). You will observe that this looks like a lens with a negative refractive index, with electrons playing the role of light rays. Materials of this nature were theorized all the way back in 1968 by Veselago. Materials of this nature are called left handed materials. An obvious application of this is to focus electrons, and it works especially well for graphene because the tunneling probability is essentially unity for small angles, so there is no dissipation as such. The refractive index can be found easily using the conservative of momentum along the y direction.

$$n \equiv \frac{\sin\phi}{\sin\theta} = -\frac{q_F}{k_F} = -\sqrt{\frac{p}{n}} \tag{3.3}$$

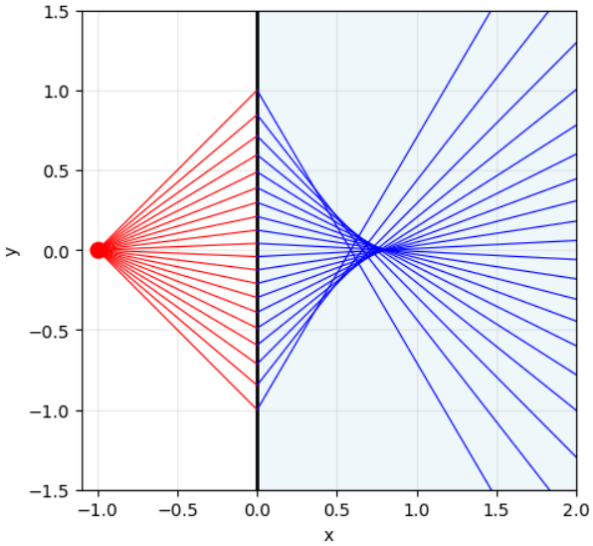


Figure 3.1: Classical trajectories of electron emanating from a point source incident on a slab of refractive index n = -0.82. The cusp is situated at x = 0.82 units.

You will notice that in Figure (3.1), there is a region on the right side where the rays are heavily concentrated and in all other regions, there is only one ray passing through. We can find the locus of those points using catastrophe theory. I shall give an intuitive explanation here without diving deep into the mathematical formalism. First, write the equation of a ray in the P region as a function of a given emanation angle ϕ.

$$y = a\tan\phi + x\tan\theta \tag{3.4}$$

where a is the distance between the point source and the junction and $\theta < 0$ is the refraction angle. To find the locus of points where more than one light ray passes through each point, we must have multiple solutions to the equation

$$f(\phi; x, y) = a\tan\phi + x\tan\theta(\phi) - y = 0 \tag{3.5}$$

where $\theta(\phi)$ is found using equation (3.3). According to catastrophe theory nomenclature, (x, y) are the control variables, the ones that you can control in an experiment. An example of a control variable would be how much force you are applying on an object in an experiment. Here, you can think of (x, y) physically as the position of a detector, whose location is in our hands. The problem is to now find the pairs (x, y) such that equation (3.5) has multiple solutions in the interval $[-\pi/2, \pi/2]$ (since those are the only rays that can reach the barrier). Take a look at Figures (3.2) and (3.3). In Figure (3.2), for the specified parameters, we only have one solution while we have three in Figures (3.3). As we change smoothly change (x, y) from $(1, 0, 5)$ to $(0.1, 0, 2)$, there must have been a critical pair of co-ordinates that divides the one solution and multiple solution

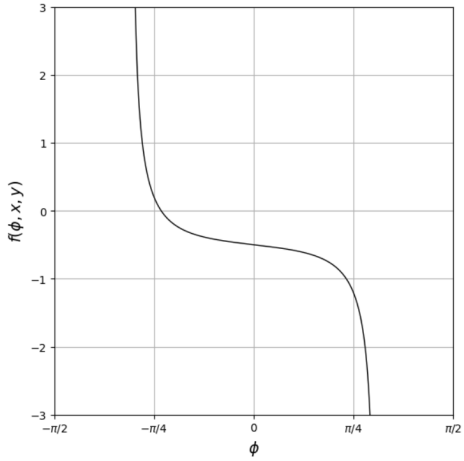


Figure 3.2: $a = 1$, $x = 1$ and $y = 0.5$

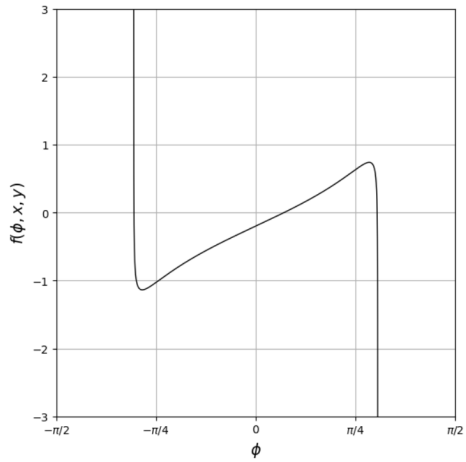


Figure 3.3: $a = 1$, $x = 0.1$ and $y = 0.2$

regimes. Given the change in shape of the function, it is not hard to guess that the derivative of f with respect to ϕ must have vanished somewhere in the middle.

$$\frac{\partial f}{\partial \phi}(\phi; x, y) = 0 \tag{3.6}$$

We can use equations (3.5) and (3.6) to eliminate ϕ to get the locus of the boundary of the this region.

$$y(x) = \pm\sqrt{\frac{(x^{2/3} - x_0^{2/3})^3}{n^2 - 1}}; x_0 = |n|a \tag{3.7}$$

The rigorous mathematical foundations of this are connected to catastrophe theory as alluded to earlier. The basic idea is that for a smooth change in some variables, a system might exhibit a discontinuous behaviour. In this case, we change the position of the detector smoothly and the discontinuity arises in the number of rays passing through the detector which jumps suddenly from one to three. However, what we've looked at so far is the classical case. We are dealing with waves, not particles. So, we must solve Dirac's equation for this case.

3.2 Quantum Mechanical Considerations

Recall Dirac's equation.

$$[i\hbar v_F \boldsymbol{\sigma} \cdot \nabla + V(x)]\psi = E\psi \tag{3.8}$$

We want to model a source that emits electrons at $\boldsymbol{a} = (-a, 0)$. To do that, we modify Dirac's equation as follows.

$$[i\hbar v_F \boldsymbol{\sigma} \cdot \nabla + V(x) - E]\psi = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \delta(\boldsymbol{r} - \boldsymbol{a}) \tag{3.9}$$

The right side represents a point source with a fixed polarization, which is the pseudospin in this case. We define Green's function, represented by $G(\boldsymbol{r}; \boldsymbol{a})$ (Green's functions are 2×2 matrices in this case) as

$$[i\hbar v_F \boldsymbol{\sigma} \cdot \nabla + V(x) - E]G(\boldsymbol{r}; \boldsymbol{a}) = \delta(\boldsymbol{r} - \boldsymbol{a}) \tag{3.10}$$

To go about solving this, one must take the fourier transform of this equation. First, let us define some parameters to non-dimensionalize this equation. The length scale of this problem is a and the energy scale is $E_0 = E$. Let's define $h = \hbar v_F / E_0 a$, $\tilde{\boldsymbol{r}} = \boldsymbol{r}/a$, $\tilde{E} = E/E_0$, $\tilde{p}_i = -ih\frac{\partial}{\partial \tilde{x}_i}$, $\tilde{V}(\tilde{x}) = V(x)/E_0$ and $\tilde{G}(\tilde{\boldsymbol{r}}; \tilde{\boldsymbol{a}}) = E_0 a^2 G(\boldsymbol{r}, \boldsymbol{a})$. With all these substitutions, we have

$$\left[\boldsymbol{\sigma} \cdot \hat{\mathbf{p}} + \tilde{V}(\tilde{x}) - \tilde{E}\right] \tilde{G}(\tilde{\mathbf{r}}; \tilde{\mathbf{a}}) = \delta(\tilde{\mathbf{r}} - \tilde{\mathbf{a}}) \tag{3.11}$$

We shall use this version of the equation and omit all the tildes to keep the equations from getting messy. Define the fourier transform as

$$f(x) = \frac{e^{i\pi/4}}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{f}(p)\, e^{ipx/\hbar}\, dp, \tag{3.12}$$

$$\tilde{f}(p) = \frac{e^{-i\pi/4}}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} f(x)\, e^{-ipx/\hbar}\, dx. \tag{3.13}$$

Taking the fourier transform of equation (3.11), we get

$$[\sigma_x \hat{p}_x + \sigma_y p_y + V(x) - E]\, \tilde{G}(x, p_y) = \frac{e^{-i\pi/4}}{\sqrt{2\pi\hbar}} \delta(x + a) \tag{3.14}$$

The method to solve this equation is to first solve the homogeneous equation, which is just the normal Dirac equation. Solving the Dirac equation without any electron source gives us two linearly independent solutions - waves incident on the barrier originating from $\pm\infty$. Let these solutions be ψ_1 (right moving incident wave) and ψ_2 (left moving incident wave). So, our trial solution is

$$\tilde{G}(x, p_y) = c_1(x)\psi_1(x) + c_2(x)\psi_2(x) \tag{3.15}$$

No waves should be incoming from $\pm\infty$. So, $c_1(-\infty) = c_2(\infty) = 0$. We can insert our ansatz in equation (3.14). Here, we will simply state the final result.

$$\tilde{G}(x, p_y) = \frac{e^{-i\pi/4}}{\sqrt{2\pi h}} \frac{i}{h} \frac{1}{2\cos\left(\frac{\phi+\theta}{2}\right)} \begin{pmatrix} e^{i\phi/2}e^{-i\theta/2} & e^{-i\phi/2}e^{-i\theta/2} \\ e^{i\phi/2}e^{i\theta/2} & e^{-i\phi/2}e^{i\theta/2} \end{pmatrix} e^{i\sqrt{E^2-p_y^2}\,a/h} e^{-i\sqrt{(E-U_0)^2-p_y^2}\,x/h} \tag{3.16}$$

We can take the inverse fourier transform of this and get the Green's function for this problem. The solution of equation (3.9) is then given by

$$\psi(\boldsymbol{r}) = G(\boldsymbol{r}; \boldsymbol{a}) \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \tag{3.17}$$

We can use this to numerically calculate the probability density of electrons emitted by a point source on the right side of the barrier. These results are numerically shown in figure (3.4) for three different polarizations.

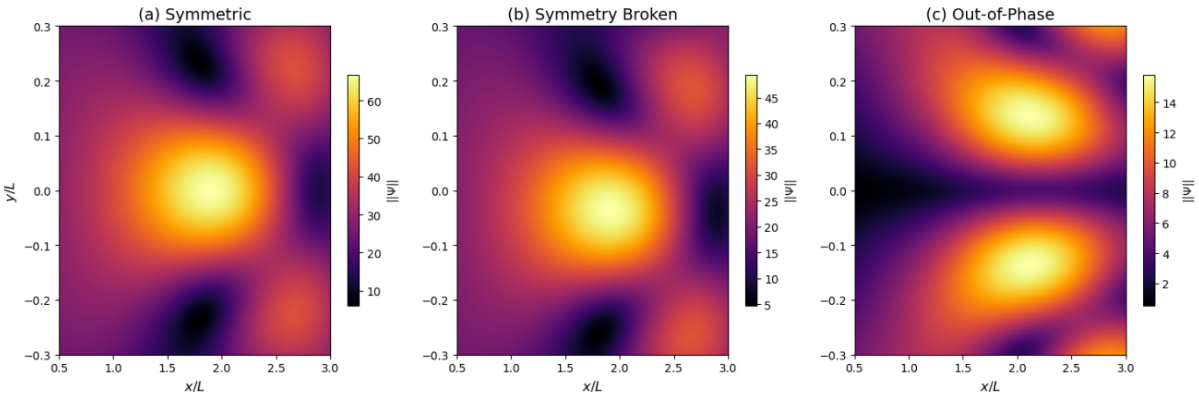


Figure 3.4: Color plots of probability density in the P side of the junction obtained through quantum mechanical calculations for 3 cases that differ by the electron psuedospin polarization. (a) $\alpha_1 = 1/\sqrt{2}$ and $\alpha_2 = 1/\sqrt{2}$, (b) $\alpha_1 = 1$ and $\alpha_2 = 0$ and (c) $\alpha_1 = 1/\sqrt{2}$ and $\alpha_2 = -1/\sqrt{2}$.

Chapter 4

Future Prospects

4.1 Veselago Lenses in Hexagonal Altermagnets

Altermagnetism is a very recently discovered paradigm of magnetism different from ferromagnetism and antiferromagnetism. Altermagnets have a spin split band structure just like ferromagnets, but have net zero magnetization just like antiferromagnets. Hexagonal altermagnets have a similar structure to graphene, so most techniques and concepts learned through this project can be applied to these materials. There is some potential for exciting new results in this area.

4.2 Tilted Dirac Cones

This is another possible area worth investigation. Some materials have a Dirac cone that is titled by some angle. This tilting bridges concepts from condensed matter and spacetime effects from relativity. Since the Dirac cone is no longer symmetric, we get to witness several anistropic electronic properties.

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